



SIMATS
ENGINEERING



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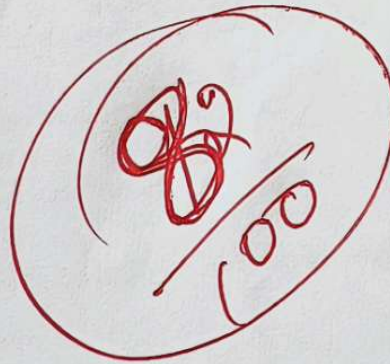
Sareetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0506- COMPUTER VISION
MODEL LAB PRACTICALS
SET 9

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Slot D.

1. Implement a face detection algorithm using Open CV to detect and locate human faces in the images.
2. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to erode using Erode function.
3. Implement a function to reverse the frames of the video to create a video in reverse mode using Open CV.
4. Implement image cropping, copying and pasting to select a region of interest (ROI) from the source image using Open CV.

va affendi



① Aim :- To implement a face detection algorithm using open cv to detect and locate human faces in the images.

Procedure :-

- Import the required open cv library
- Read the input image
- Convert the image into grayscale
- Load Haar Cascade face detector
- Detect faces using
- Draw rectangles around the detected faces
- Display the output image.

Code :-

```
img = cv2.imread("faces.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades +
    "haarcascade_frontalface_default.xml"
)

faces = face_cascade.detectMultiScale(
    gray,
    scaleFactor = 1.1,
    minNeighbors = 5,
    minSize = (30, 30)
)
```

for (x, y, w, h) in faces :

cv2.rectangle (img, (x, y), (x+w, y+h), (255, 0, 0), 2).


```
Print ("Number of faces detected: ", len(faces))
```

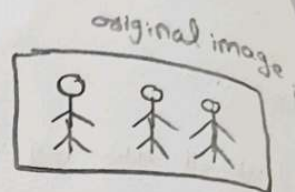
```
cv2.imshow ("Face detection", img)
```

```
cv2.waitKey (0)
```

```
cv2.destroyAllWindows ()
```

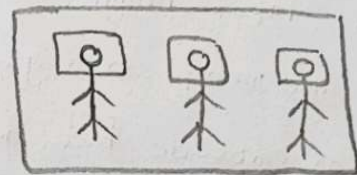
Sample input :-

faces.jpg containing one or more human faces.



Sample output :-

Number of faces detected = 3.



detected image

Result:-

Thus, the program was executed successfully.

2) Aim :- To perform basic image handling operations in Python and convert an image to an encoded image using the open cv encode function.

Procedure :-

- Import open cv and Numpy
- Read the input image
- Check whether the image is successfully loaded
- Create a structuring in kernel.

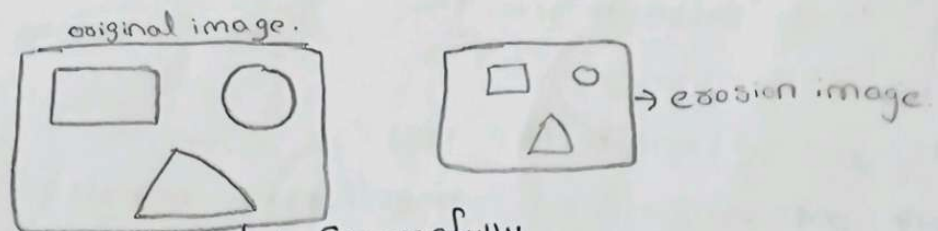
Apply the encode operation using cv2.encode ()

Display the original and encoded image.

Code:-

```
import cv2
import numpy as np
img = cv2.imread("input.jpg")
if img is None:
    print("Image not found")
    exit()
kernel = np.ones((5,5), np.uint8)
eroded = cv2.erode(img, kernel, iterations=1)
cv2.imshow("original image", img)
cv2.imshow("eroded image", eroded)
cv2.imwrite("eroded.jpg", eroded)
cv2.waitKey(0)
cv2.destroyAllWindows()
print("Erosion Completed Successfully")
```

Sample input:-



Sample output:- Erosion Completed Successfully.

The output image eroded.jpg appears with the foreground objects smaller/thinner than the original image.

Result:-

Thus, the program was executed successfully.

:- To implement a function to reverse the frames of a video create a video in reverse mode using opencv.

Procedure :-

- Import the opencv library
- Open the input video using video capture()
- Read all the frames from the video
- Stores the frames in a list
- Reverse the list of frames.
- Create a video writer object.
- Write the reversed frames into a new video
- Save the resulting video.

Code :-

```
import cv2
cap = cv2.VideoCapture("input.mp4")
if not cap.isOpened():
    print("unable to open video")
    exit()
fps = cap.get(cv2.CAP_PROP_FPS)
width = int(cap.get((cv2.CAP_PROP_FRAME_WIDTH)))
height = int(cap.get((cv2.CAP_PROP_FRAME_HEIGHT)))
frames = []
while True:
    ret, frame = cap.read()
    if not ret:
        break
    frames.append(frame)
cap.release()
frames.reverse()
```

```

fourcc = cv2.VideoWriter_fourcc(*"mp4v")
out = cv2.VideoWriter(
    "reverse_video.mp4",
    fourcc,
    fps,
    (width, height)
)

```

for frame in frames:

out.write(frame)

out.release()

Print("video reversed Successfully")

Print("Total frames:", len(frames)).

Sample input:

input. mp4 Containing a video Sequence.

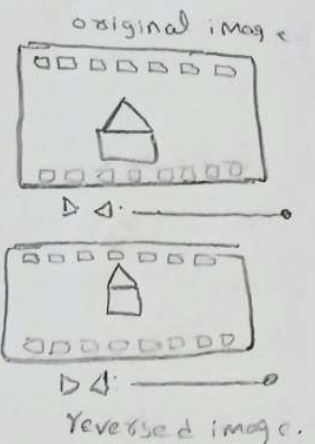
Sample output:

video reversed Successfully

Total frames = 150

Result:-

Thus, the Program was executed successfully.



Aim:- To implement image cropping, copying and Pasting operations by selecting a Region of interest Source image using open cv.

Procedure:-

import open cv.

Read the Source image

Select the required region of interest

Crop the selected region